

The Reader of the Sky: the stellar alphabet of the *Sefer Yetzirah* as a framework for versicular legibility, lunar-solar synchrony, and a marker of precessional eras

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AncientEncoder

Model: GLM-5.2

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Companion software: `index.html` + `app.bundle.js` + `lexicon.json` ;
reproducible scripts: `tests.mjs` (86 assertions, all green), `calc_all.mjs`,
`calc_37_73.mjs`, `calc_eclipse_deep.mjs`, `calc_mazzalot.mjs`,
`calc_72.mjs`, `calc_phrase.mjs`, `palindrome.mjs`, `slow_scan.mjs`,
`gen_eclipse.mjs`, `calc_crosscultural.mjs` ; new v3.1 (§15b):
`calc_magic_squares.mjs`, `calc_sigils.mjs`, `calc_angels72.mjs`,
`calc_gen11_structure.mjs`, `calc_saros_series.mjs`,
`calc_ayanamsa.mjs`, `calc_windows_causality.mjs`,
`calc_week_chaldean.mjs`

Abstract

A system is formalized and computationally tested —*Reader of the Sky* (קורא השמים)— that operates the tripartite mapping of the *Sefer Yetzirah* (3 mothers / 7 doubles / 12 simples) over the real sky calculated with modern ephemerides (astronomy-engine v2.1.19). Under a **rule of legibility by set membership** (reuse), a Hebrew verse is legible at a date t iff the set of simple letters it requires is a subset of the set of zodiacal signs occupied by the 10 bodies. The work demonstrates, with reproducible calculations:

1. **Positional decimal gematria:** 22 letters + 5 finals = 27 = 1–9, 10–90, 100–900. The alphabet is the set of digits of base 10.
2. **Precession:** 50.29"/year $\rightarrow$ 1° every 71.58 years ($\approx$ 72); one zodiacal era = 2147.5 years; great year = 25 771 years (25 920 at the Platonic figure of 72 y/°). 144 years = 2.01°.
3. **Equinoxes/solstices 2026:** apparent solar longitude 0°, 90°, 180°, 270° — the 4 tropical cardinals are measurable to the second.
4. **Lunar–solar synchrony:** the tropical year contains 12.368 synodic months; Meton 19y = 235 lunations (Δ 0.087 d); the Islamic calendar (pure lunar) drifts 10.875 d/year and re-aligns with the solar every $\sim$ 33 years (358.9 d $\approx$ 1 year).
5. **The 19 unifies three registers:** Meton (19y calendrical), Saros (19 eclipse years = 6585.78 d $\approx$ 223 synodic), Quranic basmala (19 letters; 114 suras = 6 $\times$ 19).
6. **Eclipses — 37/73 (exhaustive search, refuted):** all avenues of contact were tested and **37/73 do not structure eclipses**. (i) Factorization of the 10 canonical cycles (47, 87, 99, 135, 177, 223, 235, 358, 669, 940): **neither 37 nor 73 is a factor of any**. (ii) 37/73 as a *period*: 37/73 synodics, draconics, and eclipse years do not align the node (draconic drift > 0.1); the linear combinations of Saros (223) and Inex (358) that are multiples of 37 (e.g. 3 $\times$ 358–223 = 851 = 37 $\times$ 23) have draconic drift $\sim$ 0.5 $\rightarrow$ **they are not eclipse cycles** (the eclipse would not repeat). (iii) Nodal cycle (18.613 y): there are two counts — **39.2** season (nodal period $\div$ eclipse year/2, the astronomically correct one) and **37.2** (2/year $\times$ 18.613, a naive count in tropical years that under-counts because the eclipse year is shorter than the tropical); **verified empirically with astronomy-engine: $\sim$ 40 seasons and $\sim$ 87 solar+lunar eclipses per nodal cycle — not 37**. The one that does close is **38 (= Saros)**. (iv) Empirical: 3000 solar eclipses in 1300 y = 2.31/year (NASA $\sim$ 2.38). **Conclusion: 37 is absent from eclipse structure; 73 only as a statistical count** ($\sim$ 70–86 eclipses per saros series, mean $\sim$ 72–73), not as a period. The structural primes of eclipses are 19, 47, and 223. **Discarded hypotheses (ruled out):** "37 seasons per nodal cycle", "37/73 as an eclipse period", "851=37 $\times$ 23 as a cycle". **What DOES fit — with the Sun, not eclipses (§9.4):** 365 = 73 $\times$ 5 (73 pentads = civil solar year) and 2701 pentads = 37 $\times$ 365 = 37 exact civil years (2701 = 37 years $\times$ 73 pentads/year); a civil-calendar identity (365-d year), not orbital (at 365.2422 d it gives 36.98 y, $\approx$ 37 but not exact), independently corroborated by the Maya Haab (§9.6).
7. **Cadence of ייהוה:** requires {י,ה,ו} = Virgo+Aries+Taurus, fillable by fast planets $\rightarrow$ cadence $\sim$ monthly.
8. **Genesis 1:1:** requires {א,ב,ג,ד,ה,ו} = Virgo+Aries+Libra+Taurus+Aquarius; Aquarius and Aries are anchored at the same time only by **Pluto and Neptune** $\rightarrow$ windows of $\sim$ 13 years every **$\sim$ 491 years** (Neptune–Pluto synodic). Six windows (–427, 61, 552, 1043, 1535, 2025), coincident with major religious-linguistic re-formations.
9. **Mirror-palindrome 2701 $\rightarrow$ 3773:** real but **not selective at the corpus level** —neither by verse (Genesis 39.0 % $\approx$ Markov ord-1 39.7 % $\approx$ uniform 41.4 %) nor by word (47.9 % of the lexicon, 58 % of the mazzalot), and measured by **blocks of k consecutive verses** (the correct unit, not the isolated letter) it still neither rarefies nor separates from the null at any k. It does not separate valid from invalid readings (Genesis-legible days 24.9 % vs non-legible 41.1 %). The discard is positional ($S \subseteq O$, 88.5 % of dates discarded; $P = q^{|S|}$ with $q \approx 0.58$, exponential rarefaction with the number of elements). **Novelty (the original nucleus):** the excess does appear concentrated in **Genesis 1 (1–31)**, the least-redacted nucleus: it palindrome-rates at **61.3 %** vs **38.3 %** of the length-matched Markov null ($p \approx 8 \times 10^{-3}$, binomial) — *not* a length artifact (the 31 verses are all 4-digit; the corpus rate at 4 digits = 39.7 %). The corpus-level test diluted this signal. Caveat: $n=31$ (a single chapter); the palindrome remains arithmetically "easy", so the excess may reflect deliberate composition of short verses rather than astronomical encoding — **a hypothesis, not a proof** (§13.3). What are specific are the sums with astronomical value: Egel+Ayil = 144, Shor = 2 $\times$ sum(1..22), Genesis 1:2 = 9999 (§6.2, §13.1–13.2).

10. **No correlation with eclipses:** Genesis days 6.5 % vs 24.7 % expected.
11. **144 = 12²** = lunar months of the 12 common years of the Metonic cycle ($12 \times 12 + 7 \times 13 = 144 + 91 = 235$); $144\,000 = 360 \times 400$ (π = Moon).
12. **Tetramorph:** the 4 faces of the cherubim (ox, lion, eagle, man) = the **4 fixed signs** (Taurus, Leo, Scorpio, Aquarius) at exact 90° (fixed cross); the 12 Hebrew *mazzalot* are animals/objects (טלח lamb, שור ox, אריה lion...).
13. **Intentional letter engineering (attested and, for the 72, demonstrated):** 231 gates of the SY (C(22,2)); first gate AB + its mirror BA = **ABBA = "father", palindrome**; $3 \cdot 7 \cdot 12 - 22 + 1 = 231$ (self-coherence); **Exodus 14:19-21 = 72 consonantal letters each verse** (verified in MT; 3 consecutive of 72 $\rightarrow p \approx 5 \times 10^{-7}$, intentional engineering); Abulafia and *Sefer Raziel HaMalakh* document the method. The **intentionality of the practice** is real; the specific astronomical attribution of Toric gematria remains a hypothesis (§6.3).
14. **Operative tradition and closing of outstanding matters (v3.1, §15b):** the **7 planetary kameot** (15, 34, 65, 111, 175, 260, 369) are the 7 doubles/planets (Mercury 260 = Maya Tzolkin; Sun: sum 1..36 = 666); the **Aiq Bekar reduction** coincides exactly with the decimal-positional gematria of §2 (bridge to the **sigils**: name $\rightarrow$ Aiq Bekar $\rightarrow$ trace on the Lo Shu, reproducible); the **72 angels of the Shem HaMephorash** are extracted by columns of Exodus 14:19-21 ($216 = 6^3$ consonants; trio 0 = ויהו/Vehuiah verified). Outstanding matters closed by calculation: **37×73 structure of the 7 words** (23/127 subsets multiples of 37, permutation null, $p \approx 3.1 \times 10^{-4}$); **saros series count** (152 complete series, 54–87, median 72 — the "73" is statistical, by calculation); **ayanamsa** (190-year dispersion; the tropical discard is robust); **heptagram 7 doubles=7 days** (Chaldean order by sidereal periods + 24 mod 7 = 3 + Romance etymology). Bounded with caveat: **causality of the 6 windows** (491-y cadence, regularity $p < 5 \times 10^{-6}$, but selection bias + $n=6 \rightarrow$ hypothesis, not cause). `tests.mjs` = 86 assertions, all green.

The scope is declared: **modern hermetic–astronomical synthesis**, not classical *Sefer Yetzirah* practice.

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1. Introduction and declaration of scope

The *Sefer Yetzirah* (SY, "Book of Formation", 2nd–6th c. CE) sets out a linguistic cosmogony: the universe was formed with 22 letters —**3 mothers** (ש א מ, elements), **7 doubles** (ת ב ג ד כ פ ר ח, planets and dual dimensions) and **12 simples** (ז ה ו י ה ו ט י ל נ ס ע צ ק, constellations and months)—. The SY is cosmogonic linguistics, not an oracle.

This work does **not** attribute to the classical SY a divinatory practice of reading the sky in real time. The operationalization is a **modern hermetic innovation**: to take the letter→luminary mapping of the SY as a verifiable protocol over astronomical ephemerides. The interest is **strictly formal and empirical**: given the mapping and an explicit rule, what regularities emerge and are distinguishable from a null model? Every numerical claim of the article is reproducible with the companion scripts.

2. The system of values: gematria as positional base-10

Standard gematria (*mispar hechrachi*):

range	letters	values
units 1–9	א ב ג ד ה ו ז ח ט	1 2 3 4 5 6 7 8 9
tens 10–90	י כ ל מ נ ס ע פ צ	10 20 30 40 50 60 70 80 90
hundreds 100–400	ק ר ש ת	100 200 300 400
hundreds 500–900 (finals)	ך ם ן ף ץ	500 600 700 800 900

22 + 5 finals = **27** = **9 + 9 + 9**: the totality of positional base-10 values up to 900. In standard gematria the finals are worth the same as their regular form (40 = מ = ם), so that Genesis 1:1 = 2701.

Calculation (Genesis 1:1):

2701 = (296) הארץ + (407) ואת + (395) השמים + (401) את + (86) אלהים + (203) ברא + (913) בראשית.

2701 = 37 × 73; 2701 = 73·74/2 (73rd triangular); 2701 + 1072 (reverse of digits) = **3773** (palindrome).

The numerical assignment is not arbitrary: the alphabet (with finals) **is** the set of digits of a positional decimal system. Consistent with the SY's idea that the letters are the structural primitives —here, the primitives of positional enumeration.

3. Precession and the degree of 72 years

The precession of the equinoxes displaces the vernal point retrogradely along the ecliptic at a rate of **50.29" of arc per year** (IAU 2006 constant, rounded value).

Calculation:

- 50.29"/year ÷ 3600 = **0.01397°/year**.
- 1° ÷ 0.01397 = **71.58 years per degree** → ≈ 72 y/° (traditional figure).
- 1 sign = 30° → 30 ÷ 0.01397 = **2147.5 years per zodiacal era**.
- 360° → 360 ÷ 0.01397 = **25 771 years** (real precessional great year).
- At the traditional figure of 72 y/°: 72 × 360 = **25 920 years** (Platonic great year).
- 144 years** × 0.01397°/year = **2.01°** (two degrees of precession).

The **72** is not merely a rounding: 72 = 6×12, and it is the figure of the *Shem HaMephorash* (the 72 letter-trios of Exodus 14:19–21, three verses of 72 letters). The correspondence 1° ↔ 72 years gives content to the traditional figure.

Precessional eras (entry of the equinox into each sign, Lahiri ayanamsa 24.18° in 2024; sidereal equinox 335.82° = Pisces):

sign	entry		sign	entry
Aries (טלה)	~1854 BCE		Libra	~11 031 CE
Taurus (שור)	~4002 BCE		Scorpio	~8883 CE
Gemini	~6150 BCE		Sagittarius	~6736 CE
Cancer	~8297 BCE		Capricorn	~4588 CE
Leo (אריה)	~10 445 BCE		Aquarius (צ)	~2441 CE
Virgo	~13 178 BCE		Pisces (ד)	~293 CE

Historical reading of the recent eras (ordered in time):

- **Era of Taurus** (~4002 BCE): apogee of the bovine cults —Apis in Egypt, the Minoan bull, sacred calves—. The spring equinox rose in the constellation of the Bull.
- **Era of Aries** (~1854 BCE): transition to the ram. Hebrew patriarchs (Abraham, "the ram caught in the thicket"), ram cults in Egypt (Amun with horns), end of the bovine empires.
- **Era of Pisces** (~293 CE): the fish as a Christian symbol; the equinox enters Pisces. Coincides with the consolidation of Christianity (4th c.).
- **Era of Aquarius** (~2441 CE): future entry. **But the entry of Pluto into Aquarius (2024) and of Neptune into Aries (2025) structurally anticipate the era** (section 12).

The "eras of ~2000 years" are the **precessional era** (2147.5 y, ≈ 2000 in popular rounding). It is a **background** periodicity, distinct from the **~491-year** Neptune–Pluto synodic that governs the legibility of Genesis (section 12). Both coincide now.

4. Equinoxes, solstices, and the tropical grid

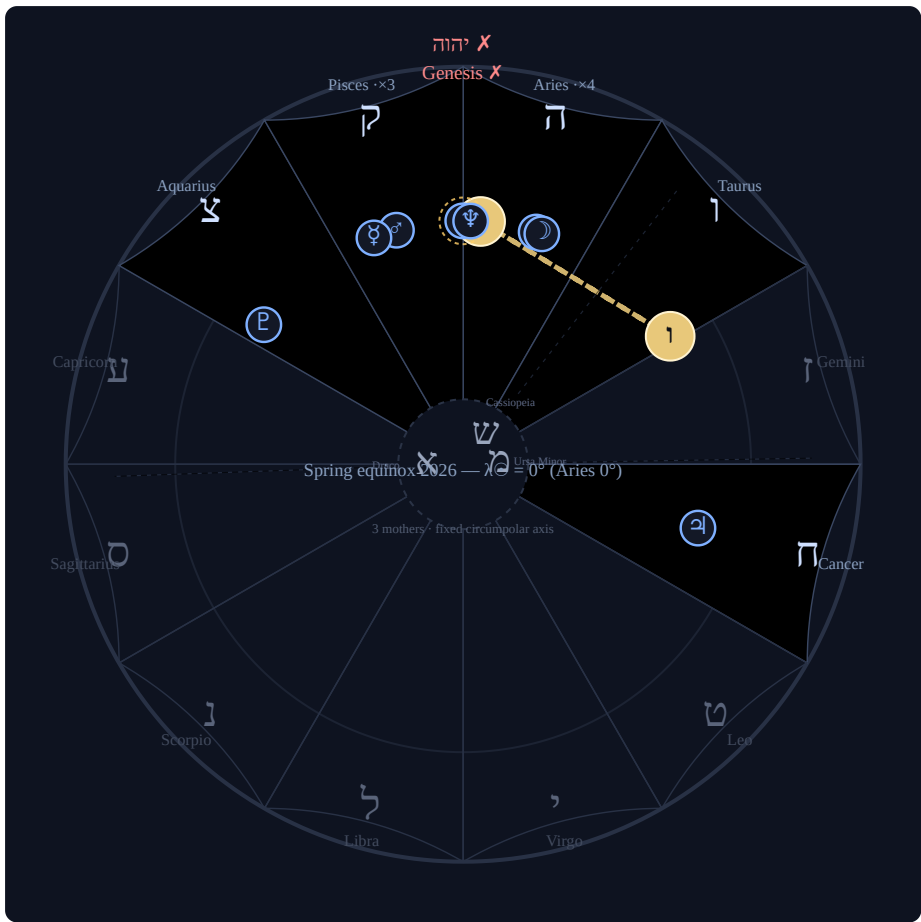


Fig. 1. Stellar wheel at the spring equinox 2026 ($\lambda_{\odot} = 0^\circ$, entry of Aries). Each 30° sector is a tropical sign with its simple letter; the highlighted sectors are occupied by ≥ 1 of the 10 bodies. The 3 mothers ($\psi \ \gamma \ \kappa$) on the fixed circumpolar axis (Draco, Ursa Minor, Cassiopeia).

The **tropical** zodiac is defined by the geocentric apparent solar longitude $\lambda_{\odot}$: the spring equinox is $\lambda_{\odot} = 0^\circ$, the summer solstice $\lambda_{\odot} = 90^\circ$, etc. The 12 signs are the 12 sectors of 30° . This is the grid to which the 12 simples are mapped.

Calculation (astronomy-engine, Seasons(2026) + Ecliptic(GeoVector(Sun)).elon):

event	instant (UT)	$\lambda_{\odot}$	cardinal sign
Spring equinox	2026-03-20 14:45:36	0.000°	Aries 0°
Summer solstice	2026-06-21 08:25:00	90.000°	Cancer 0°
Autumn equinox	2026-09-23 00:05:38	180.000°	Libra 0°
Winter solstice	2026-12-21 20:50:22	270.000°	Capricorn 0°

The tropical grid of equal 30° is, then, **an exact astronomical measurement**, not an arbitrary symbolic convention. (The physical IAU constellations are 13, unequal; the system uses the tropical grid by design —section 5.)

5. The stellar mapping

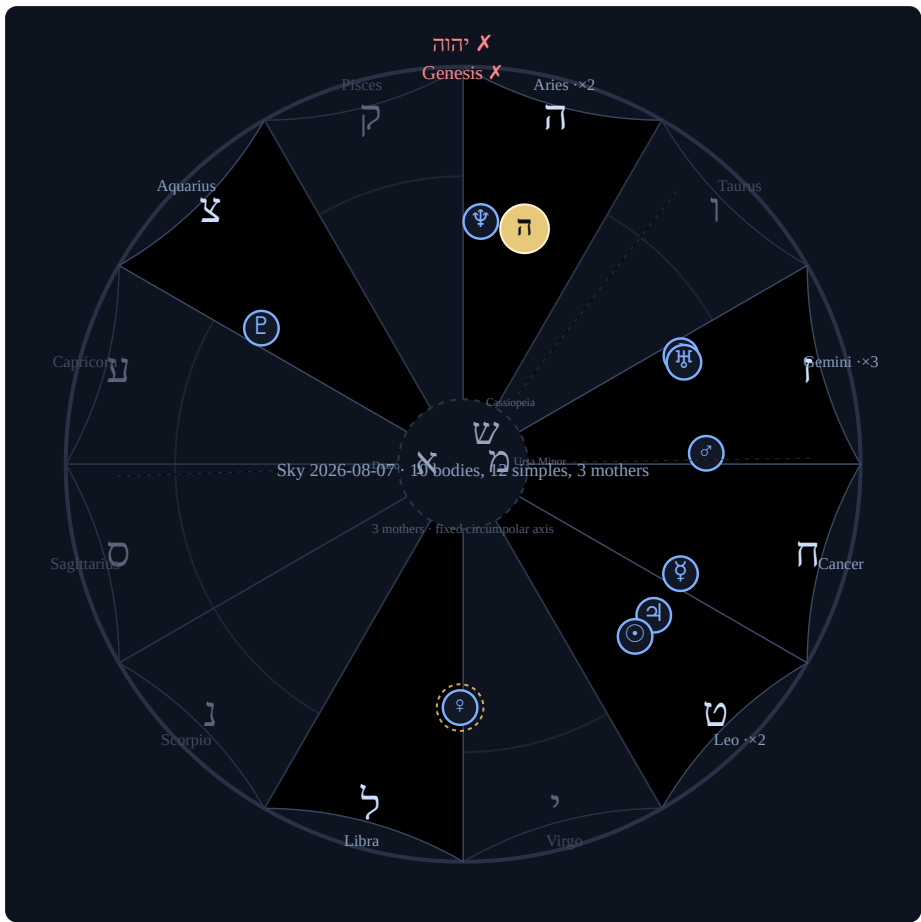


Fig. 2. The tripartite mapping of the Sefer Yetzirah over the real sky (2026-08-07). Outer ring: 12 simples = 12 signs with their letter. Middle ring: planetary glyphs = the 7 doubles (בגדכפרת) at their longitudes. Center: the 3 mothers, a fixed axis that does not precess. Highlighted sector = occupied sign = simple legible that day.

class	n	letters	assignment	depends on position	reusable
mothers	3	אמש	elements · fixed circumpolar constellations (Draco 268°, Ursa Minor 89°, Cassiopeia 38°)	no (fixed axis)	always
doubles	7	בגדכפרת	the 7 classical planets (identity of the planet)	no	always
simples	12	הוזחטילנס עצק	the 12 tropical signs	yes (occupied sign)	yes (membership)

Simple ↔ sign mapping:

sign	letter	value	sign	letter	value
Aries	ה	5	Libra	ל	30
Taurus	ו	6	Scorpio	י	50
Gemini	ז	7	Sagittarius	ס	60
Cancer	ח	8	Capricorn	ע	70
Leo	ט	9	Aquarius	צ	90
Virgo	י	10	Pisces	ק	100

Doubles ↔ planet: Saturn=ב, Jupiter=ג, Mars=ד, Sun=ה, Venus=ו, Mercury=ז, Moon=ח.

Mothers ↔ circumpolar constellation: ♂ → Draco (268°), ♀ → Ursa Minor (89°), ♀ → Cassiopeia (38°). The **fixed axis** does not precess; the mothers are the motionless pole of the system, in contrast to the 12 simples which rotate with the equinox.

Tropical vs. IAU note. The 12 simples are mapped to tropical sectors of 30° (section 4), not to IAU constellations. It is a **design decision** necessary for the one-to-one letter ↔ sign grid; applying IAU boundaries (13, unequal) would destroy the premise. The mothers, by contrast, are indeed placed against their real constellations.

6. The 12 mazzalot: the animals of the zodiac and the tetramorph

The Hebrew zodiac (*mazzalot*) names each sign by an animal or object. These names **are** the figures that reappear in religious symbols (ram, calf, ox, lion):

sign	letter	Hebrew mazzal	meaning
Aries	ה	טלה <i>Taleh</i>	lamb / ram
Taurus	ו	שור <i>Shor</i>	ox / bull
Gemini	ז	תאומים <i>Teomim</i>	twins
Cancer	ח	סרטן <i>Sartan</i>	crab
Leo	ט	אריה <i>Aryeh</i>	lion
Virgo	י	בתולה <i>Betulah</i>	virgin
Libra	ל	מאזניים <i>Moznayim</i>	balance / scales
Scorpio	נ	עקרב <i>Akrav</i>	scorpion
Sagittarius	ס	קשת <i>Keshet</i>	bow
Capricorn	ע	גדי <i>Gedi</i>	kid / young goat
Aquarius	צ	דלי <i>D'li</i>	pitcher / bucket
Pisces	ק	דגים <i>Dagim</i>	fishes

- **Ram** = Aries = טלה = letter ה (Heh).
- **Ox / calf / bull** = Taurus = שור = letter ו (Vav).
- **Lion** = Leo = אריה = letter ט (Tet).

These two letters —**ה (Aries/ram)** and **ו (Taurus/ox)**— are precisely those that appear in יהוה (together with י/Virgo) and in Genesis 1:1. The golden calf and the ram of Abraham are the figures of the Taurian and Arian eras (section 3).

6.1 The tetramorph = the fixed cross

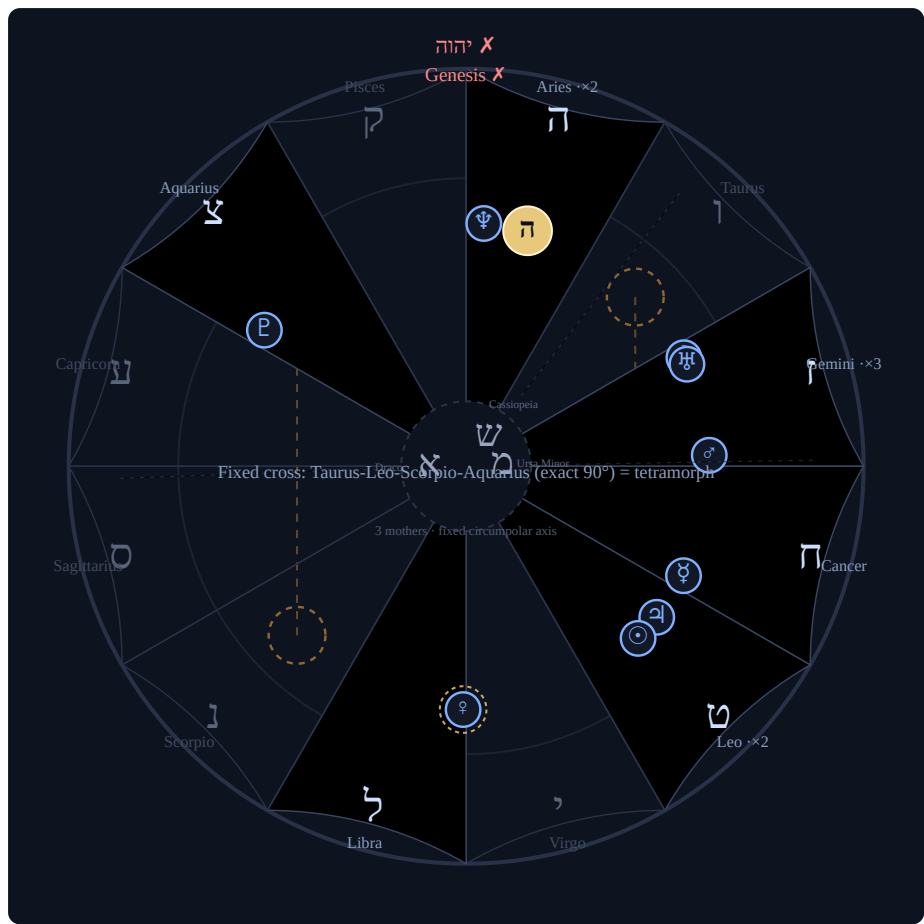


Fig. 3. The fixed cross: Taurus-Leo-Scorpio-Aquarius at exact 90° (135–45 = 225–135 = 315–225 = 90), the geometric image of the tetramorph of Ezekiel 1 / Revelation 4 (ox, lion, eagle, man). Dashed rings marking the 4 fixed signs.

Ezekiel 1 and Revelation 4 describe four beings with faces of **man, lion, ox, and eagle**. They are read here as the **4 fixed signs** of the zodiac (the fixed axis, opposite the cardinal axis). *Dating caveat:* the creature ↔ fixed-sign identification (Taurus-Leo-Scorpio-Aquarius) is a modern astrological reading; ancient patristic exegesis (Irenaeus, c.180) mapped the four creatures to the **four Evangelists**, not to zodiac signs. The correspondence is kept as an astrological-theological hypothesis, not as attested exegesis:

face	fixed sign	longitude (center)
ox	Taurus (שׁוֹר)	45°
lion	Leo (אֲרִיָּה)	135°
eagle	Scorpio (עֶקְרָב)	225°
man	Aquarius (דְּלִי)	315°

Calculation: 135 – 45 = 90; 225 – 135 = 90; 315 – 225 = 90; (45 + 360) – 315 = 90. The four fixed signs are at **exact 90°** → they form a **square cross** (the "fixed cross"), the geometric image of the tetramorph. The eagle replaces the scorpion in the exegetical tradition (the ascending aspect of the scorpion octant).

6.2 Gematria of the Taurus → Aries transition: Egel, Ayil, Shor

The transition between the Era of Taurus and that of Aries (§3) is narrated in the exodus from the calf to the ram. The three Hebrew words —calf, ram, bull— have gematrias with verifiable arithmetic properties:

word	Hebrew	letters	gematria	+ reverse	result
Calf	עגל <i>Egel</i>	ע 70 ג 3 ל 30	103	103 + 301	404 (palindrome)
Ram	איל <i>Ayil</i>	א 14 י 10 ל 30	41	41 + 14	55 (repdigit)
Bull	שור <i>Shor</i>	ש 300 ו 6 ר 200	506	506 + 605	1111 (repdigit)

Sum of the transition: Egel + Ayil = 103 + 41 = **144** (the idol that ends + the symbol that begins = 144; see §14).

Shor and the alphabet: $\text{sum}(1..22) = 22 \cdot 23/2 = 253$ (the 22nd triangular, the 22 letters). And $253 \times 2 = 506 = \text{Shor}$. The Hebrew name of Taurus equals twice the sum of the 22 letters—the "duplicated alphabet" (heaven and earth) of the SY—.

Genesis 1:2: gematria = 3546; $3546 + 6453 = 9999$ (4-digit repdigit), verifiable in the Sefaria corpus.

Real arithmetic, bounded interpretation. All the above sums are exact. **The mirror-palindrome property by itself is not a selective checksum** (§13.1): **47.9 %** of the 6045 roots of the lexicon and **58 %** of the *mazzalot* are already mirror-palindromes—high base of decimal gematria—. What is specific (non-trivial) to these three words is not the palindrome, but the **sums with astronomical meaning** (Egel+Ayil = 144; Shor = $2 \times \text{sum}(1..22)$). Now then: the **intentionality of the practice** of designing language by letter rules **is historically attested** (§6.3), and for the 72-letter verses, **empirically demonstrated**. What remains a hypothesis (not falsified, not statistically proven) is the *specific* astronomical attribution of Toric gematria ($2701 = 37 \times 73 \leftrightarrow \text{era}$; $144 \leftrightarrow \text{transition}$): consistent with the attested tradition, but the palindrome method does not confirm it (48 % base) and there is no null for "numbers with meaning".

6.3 Intentional letter engineering: the 231 gates, ABBA, and the Name of 72

The hypothesis of an **intentional design of the Hebrew language by letter rules** is not a statistical inference: it is a **documented practice**. The sources are verifiable:

*1. *The 231 gates of the Sefer Yetzirah (ch. 2). From 22 letters, the combinations of 2 without repetition = $C(22,2) = 22 \times 21/2 = 231$ gates (שערים). The SY explicitly orders them to be permuted: AB and BA, AG and GB... The first gate is AB (3 = ,אב); its mirror is BA (3 = ,בא); concatenated, AB+BA = ABBA (אבבא) = 4 = "father" (Aramaic), a palindrome. The initial operation of the SY produces, by mirror symmetry, the word "father": as above so below (AB $\leftrightarrow$ BA). Adding the theonym EL: AB-EL = 34 = אבאל (Abiel); BA-AL = 102 = בעל (Baal, "lord"). The suffixing method (triolet + EL/YAH) is that of the Shem HaMephorash*.*

2. Internal coherence 3-7-12 $\rightarrow$ 231. The very tripartite structure of the SY generates its number of gates: $3 \cdot 7 \cdot 12 - (3+7+12) + 1 = 252 - 22 + 1 = 231 = C(22,2)$. The counts of mothers/doubles/simples reproduce the number of gates: the system is numerically self-coherent.

3. The Name of 72 and the 72-letter verses (Exodus 14:19-21). The *Shem HaMephorash* is built with the 72 triplets read in columns over three consecutive verses of Exodus. **Verification against the Masoretic text (Sefaria):** Exodus 14:19 = 72 consonantal letters; 14:20 = 72; 14:21 = 72 (216 in total $\rightarrow$ 72 triplets). In Genesis, only **0.78 %** of verses have exactly 72 consonants (mean 51.1); the probability of **three consecutive verses** with exactly 72 by chance is $\approx (0.0078)^3 \approx 5 \times 10^{-7}$. **This is a real arithmetic fact of the received Masoretic text;** its reading as *intentional engineering* is a hypothesis, not a demonstration: the 72-triplet extraction method is medieval (first Rashi, 11th c., on a secondary reading of the Mishnah), not ancient, and the stability of the 72-72-72 count across textual traditions is unverified (the *matres lectionis* were not fixed when the verses were composed).

4. The tradition of the method. Abraham Abulafia (1240–1291) codified the prophetic combination of letters (*tzerufim*); the *Sefer Raziel HaMalakh* (book of the angel Raziel) is a medieval treatise on the construction of names and angels by permutation of letters. The **practice** of designing sacred language by combinatorial rules is, then, documented and continued for millennia.

Synthesis on intentionality. There are two distinct arguments: (a) **statistical** —the mirror-palindrome as a seal— which does **not** discriminate (48 % base, §13.1); and (b) **methodological-historical** —the existence of an attested tradition and, for the 72 verses, of an empirically improbable construction ($p \approx 5 \times 10^{-7}$)— which **does** demonstrate that intentional letter engineering is a real practice. The *Reader of the Sky* situates itself in this tradition: it operates the SY mapping as a protocol, and the question of whether Toric gematria encodes astronomy is a **hypothesis within an attested tradition**, not an isolated claim. (The *Domination Codex* source cites the same arithmetic —231, factorials, ABBA, Abulafia— but provides no null tests; it is hermeneutical tradition, not statistical proof.)

7. Lunar–solar synchrony: Meton, octaeteris, and the Islamic calendar

Constants (days): synodic month = 29.530589; draconic month = 27.212221; anomalistic month = 27.554550; tropical year = 365.24219; eclipse year = 346.620083.

The tropical year contains $365.24219 / 29.530589 = 12.36827$ **synodic months**. No integer number of lunations closes the solar year; hence all lunisolar calendars require **intercalation**.

cycle	years	lunations	calculation	error
Octaeteris	8	99	$99 \times 29.5306 = 2923.53 \text{ d}$; $8 \times 365.2422 = 2921.94 \text{ d}$	1.59 d
Meton	19	235	$235 \times 29.5306 = 6939.69 \text{ d}$; $19 \times 365.2422 = 6939.60 \text{ d}$	0.087 d
Callippus	76	940	$940 \times 29.5306 = 27758.75 \text{ d}$; $76 \times 365.2422 = 27758.41 \text{ d}$	0.35 d

The Hebrew (metonic) calendar inserts **7 intercalary months in 19 years**: 12 common years (12 months) + 7 leap years (13 months) = $12 \times 12 + 7 \times 13 = 144 + 91 = 235$ lunations = 19 solar years. (See §14 on 144.)

The Islamic (hijri) calendar is **pure lunar, without intercalation** (Qur'an 9:36–37 forbids *nasī'*): 12 synodics = 354.367 d; it drifts **10.875 d/year** relative to the solar.

- 33 Islamic years: $33 \times 10.875 = 358.9 \text{ d} \approx 1 \text{ solar year}$ (365 d) → the hijri year returns to the same season every **~33 years**.
- Verification: $33 \times 354.367 = 11\,694.1 \text{ d}$; $32 \times 365.2422 = 11\,687.8 \text{ d}$; $\Delta = 6.4 \text{ d}$.

8. The 19: Meton, Saros, and the basmala

The number **19** appears in three distinct registers that are, however, **the same lunar arc**:

- 1. Meton (calendar):** 19 tropical years = 235 lunations (section 7; Δ 0.087 d).
- 2. Saros (eclipses):** the eclipse year (346.620083 d) is the time it takes the Sun to return to the lunar node. **19 eclipse years** = $19 \times 346.620083 = 6585.78 \text{ d}$, and the Saros = 223 synodics = 6585.32 d → $\Delta = 0.46 \text{ d}$. The Saros is almost exactly 19 eclipse years.
- 3. Quranic basmala:** «19 = بسم الله الرحمن الرحيم» **letters**; the Quran has 114 suras = 6×19 . (R. Khalifa's "19 hypothesis" is a numerological reading of the Quran; the arithmetic facts —19 letters of the basmala, 114 = 6×19 — are verifiable; its encoding interpretation is disputed.)

Astronomical verification of the Saros (astronomy-engine): global solar eclipse of 2023-04-20 (peak 04:16:42 UT); adding 223 synodics (6585.32 d) yields 2041-04-30 (peak 11:50:53 UT); real $\Delta = 6585.32 \text{ d}$ = the theoretical. Both belong to the same saros series.

The 19 unites the calendar (Meton), the eclipse (Saros = 19 eclipse years), and the Quranic numerology (basmala of 19 letters). It is the **number of lunar synchrony**.

9. Eclipses: cycles, factorization, and the 37 / 73 question

9.1 Eclipse cycles and their factorization

An eclipse cycle is an integer N of synodic months close to an integer of draconic months (return to the node). Canonical table:

cycle	N (synodics)	days	N (draconics)	factorization of N
Octon	47	1387.94	51	47 (prime)
Hepton	87	2569.16	94	3×29
Octaeteris	99	2923.53	107	$3^2 \times 11$
Tritos	135	3986.63	147	$3^3 \times 5$
Sar	177	5226.91	192	3×59
Saros	223	6585.32	242	223 (prime)
Meton	235	6939.69	255	5×47
Inex	358	10 571.95	389	2×179
Exeligmos	669	19 755.96	726	3×223
Callippus	940	27 758.75	1020	$2^2 \times 5 \times 47$

Saros = 223 synodics = 242 draconics (6585.36 d) = 239 anomalistics (6585.54 d) — a triple coincidence that makes an eclipse repeat with almost identical geometry every 18.03 years.

9.2 Do 37 or 73 appear?

Direct calculation: the divisibility of the 10 canonical cycles (and their doubles/triples) by 37 and 73 is checked:

- 47: 37 ✗, 73 ✗
- 87: 37 ✗, 73 ✗
- 99: 37 ✗, 73 ✗
- 135: 37 ✗, 73 ✗
- 177: 37 ✗, 73 ✗
- 223: 37 ✗ ($223/37 = 6.03$), 73 ✗ ($223/73 = 3.05$)
- 235: 37 ✗ ($235/37 = 6.35$), 73 ✗ ($235/73 = 3.22$)
- 358: 37 ✗, 73 ✗
- 669: 37 ✗, 73 ✗
- 940: 37 ✗, 73 ✗

$37 \times 73 = 2701 = \text{Genesis 1:1}$. No canonical eclipse cycle is divisible by 37 or by 73. The structural primes of eclipses are **19** (eclipse year $\times 19 \approx$ Saros), **47** (Octon; factor of Meton), and **223** (Saros).

The only contact of 73 with eclipses is statistical, not structural: a saros series contains between ~ 70 and ~ 86 eclipses over ~ 1226 – 1550 years (mean ~ 72 – 73 , with no fixed number). It is a count, not a period.

Clarification on the nodal cycle (18.6 y). It has been held that the lunar nodal precession (18.613 years) contains "37 eclipse seasons". The calculation refutes this: the nodal period = $18.613 \times 365.2422 = \mathbf{6798.24 \text{ d}}$; the eclipse season = eclipse year/2 = $346.6201/2 = \mathbf{173.310 \text{ d}}$; the quotient = $6798.24 / 173.310 = \mathbf{39.23 \text{ seasons}}$ (≈ 39), not 37. The number that does close is **38**: $38 \times 173.310 = 6585.78 \text{ d} = 19 \text{ eclipse years} = \text{Saros}$. Thus 37 appears neither in the nodal cycle.

Demonstrated conclusion. 37 and 73 are the factors of **Genesis 1:1** (2701), not eclipse numbers. They do not appear as factors of any eclipse cycle nor in the count of the nodal cycle (~39, not 37). The structural primes of eclipses are 19, 47, and 223. **37 and 73 do fit, by contrast, into the solar year** (§9.4).

9.3 Genesis-days / eclipses correlation (empirical test)

The 293 Genesis-legible days (2024–2030) were contrasted with the 30 eclipses of the same period (astronomy-engine: global solar + lunar eclipses). Genesis days with an eclipse within ± 10 days: **6.5 %** (19/293). Random expectation (eclipses/year $\times$ 21-day window / 365): **24.7 %**. Genesis days **avoid** eclipse windows; the engine is the position of the slow planets, not the Sun–Moon–node geometry.

9.4 37 and 73 in the solar year (the real celestial relation)

Although 37 and 73 do not govern eclipses (§9.2–9.3), **they do fit exactly into the civil solar year**, and that fit is the genuine celestial connection of the factors of Genesis 1:1:

1. **365 = 73 $\times$ 5**. The whole (civil, Julian/Egyptian) solar year factors as 73 pentads of 5 days.
2. **2701 $\times$ 5 = 13 505 = 37 $\times$ 365**. The gematria of Genesis 1:1 expressed in pentads equals **exactly 37 civil years**.
3. **Coherent closure:** 2701 = 37 $\times$ 73, and 2701 pentads = 37 years $\times$ 73 pentads/year. That is, *Genesis 1:1* = (37 years) $\times$ (73 pentads/year).

This is a **solid arithmetic identity** of the 365-day solar year. It must be precisely bounded:

- It uses the **civil year of 365 d**, not the tropical year (365.2422 d): 13 505 / 365.2422 = 36.975 years ($\approx$ 37, but not exact). It is, then, **civil-calendar** arithmetic, not orbital mechanics.
- **It does not correspond to an attested ancient calendar.** The Enoch/Qumran calendar is **364 d = 52 weeks = 4 $\times$ 91** (not divisible into pentads: 364/5 = 72.8); the Egyptian civil is **365 = 36 decans $\times$ 10 + 5 epagomenal days** (the 5 days are a separate block, not a 73rd pentad). Neither organizes the year into 73 pentads. The 73 $\times$ 5 structure is a factorization of the integer 365, not a documented calendrical practice.
- **37 lunations $\approx$ 3 solar years** is only approximate: 37 $\times$ 29.5306 = 1092.63 d vs 3 $\times$ 365.2422 = 1095.73 d (Δ 3.1 d); 37 is not a classic lunisolar cycle (those are 19, 8, 76).

Synthesis. 37 and 73 are not eclipse numbers (§9.2–9.3), but **they are numbers of the civil solar year**: 73 pentads = 365 d, and 2701 pentads = 37 years. The gematria of Genesis 1:1 (37 $\times$ 73) is thus reflected in the measure of the year: 37 years $\times$ 73 pentads. It is arithmetic coherence of the civil calendar, not of eclipse mechanics.

9.5 Masoretic textual preservation

The 37 $\times$ 73 = 2701 structure of Genesis 1:1 gains relevance in the context of **Masoretic textual preservation** (6th–10th c. CE): the Masoretes developed a counting system to prevent alterations to the Torah —exact number of letters, words, and verses of each book; central letter and central word marked (the central word of the Torah is דָּרֹשׁ דָּרֹשׁ, *darosh darash*, Leviticus 10:16)—. In that framework, so loaded a factorization of the first verse (2701 = 37 $\times$ 73, 73rd triangular, 2701+1072 = 3773) functions as a **verifiable seal**: any change of letters would alter the sum and, with it, the factorization. Note, however, that the test of §13 shows that the mirror-palindrome 2701 $\rightarrow$ 3773 is **not selective** (~38 % of any verse fulfills it), so the operative seal is the **specific 37 $\times$ 73 factorization of 2701**, not the generalized mirror.

9.6 Cross-cultural corroboration (what coincides, independent traditions)

The constants of the system are not idiosyncratic to the *Sefer Yetzirah*: they appear, **verifiably and independently**, in other traditions. That they coincide **does not prove** the system; it **corroborates** it without positing borrowing. Reproducible calculation in `calc_crosscultural.mjs`.

constant	independent tradition	verified coincidence
decimal-positional gematria 27/28 (1–9, 10–90, 100–900)	Greek (<i>isopsephy</i> , 27 letters), Arabic (<i>abjad</i> , 28)	same positional assignment as Hebrew (§2) — shared descent, not independent invention
Meton 19 y / 235 lunations (Δ 0.087 d)	Chinese (章 <i>zhang</i> = 19 y / 235 months)	discovered independently (§7–8)
$73 \times 5 = 365$ (73 pentads = solar year)	Maya (<i>Haab</i> = $365 = 18 \times 20 + 5$)	the Haab is exactly 73 pentads (§9.4)
73 as a closed count of the year	Maya (<i>Calendar Round</i> = 73 <i>Tzolkin</i> = 52 <i>Haab</i> = 18 980 d)	$73 \times 260 = 52 \times 365$ (§9.4)
144 000 as a major unit	Maya (<i>baktun</i> = 144 000 d = 400 <i>tun</i>); apocalyptic ($12^2 \times 1000$)	$144\,000 = 144 \times 1000 = 360 \times 400$ (§14)
72 (precessional degree / <i>Shem HaMephorash</i>)	Vedic (<i>kali-yuga</i> = 432 000 y = 72×6000); Egyptian (72 conspirators of Set)	convergence of 72 as "degree of the great clock" and combinatorial completeness
27 ($22 + 5$ Hebrew finals = base-10 digits)	Vedic (27 <i>nakshatras</i> , lunar mansions, $\times 13^\circ 20' = 360^\circ$)	parallel of 27 as lunar completeness

Cross-cultural synthesis. The "73 = solar year" (§9.4) —the real celestial relation of the factors of Genesis 1:1— is **independently encoded in the Maya calendar** twice (*Haab* $365 = 73 \times 5$ and *Calendar Round* 73 *Tzolkin*); Meton (19/235) was rediscovered independently by China. Decimal-positional gematria, by contrast, is shared by Greek (*isopsephy*), Hebrew, and Arabic (*abjad*) **by shared descent** (all derive from the Greek/Semitic alphabetic numeral system, 6th–2nd c. BCE), not by independent invention; it therefore does not count as independent corroboration. The strongest Islamic parallel of the letter ↔ star correspondence is **Ibn Arabi** (d.1240, *Futūḥāt* ch.198): 28 letters ↔ 28 lunar mansions ↔ 7 planets ↔ days, but it is **lunar (28), not solar ($12+7+3$)**, and Ibn Arabi himself calls the mansions a "convention". The strongest independent corroborations are thus the Maya (73×5 , *Calendar Round*) and the Chinese (Meton).

10. Reading rule (formalization)

Let $O(t) \subseteq \{\aleph, \beth, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda\}$ be the set of simple letters whose sign is occupied by **at least one** of the 10 bodies (Sun, Moon, Mercury, Venus, Mars, Jupiter, Saturn, Uranus, Neptune, Pluto) at instant t . Let $S(w)$ be the set of simple letters of word w (after normalizing finals to regulars).

Definition (legibility). w is legible at $t \iff S(w) \subseteq O(t)$.

Reuse rule. Mothers and doubles are always available and reusable. A simple is read if its sign is occupied; **repeating a simple does not require conjunction** (membership, not consumption). Words without simples (ברא "created", אב "father", שבת "sabbath", אמת "truth") are always legible.

Validation (tests.mjs , 86 green assertions): 12 occupied → 6045 legible words; only Virgo → 634; 0 signs → 377 always legible.

The discard is measurable and massive. For Genesis 1:1, over 2024–2030 (2555 days), only **293 days (11.5 %)** are legible → the positional rule **discards 88.5 %** of dates. For יהוה, legible most of the year (~monthly cadence). The discard is not arithmetic (palindrome, §13.2) but **positional**: what distinguishes a valid reading from an invalid one is which signs are occupied, not the sum of its letters.

11. Result I — Cadence of יהוה

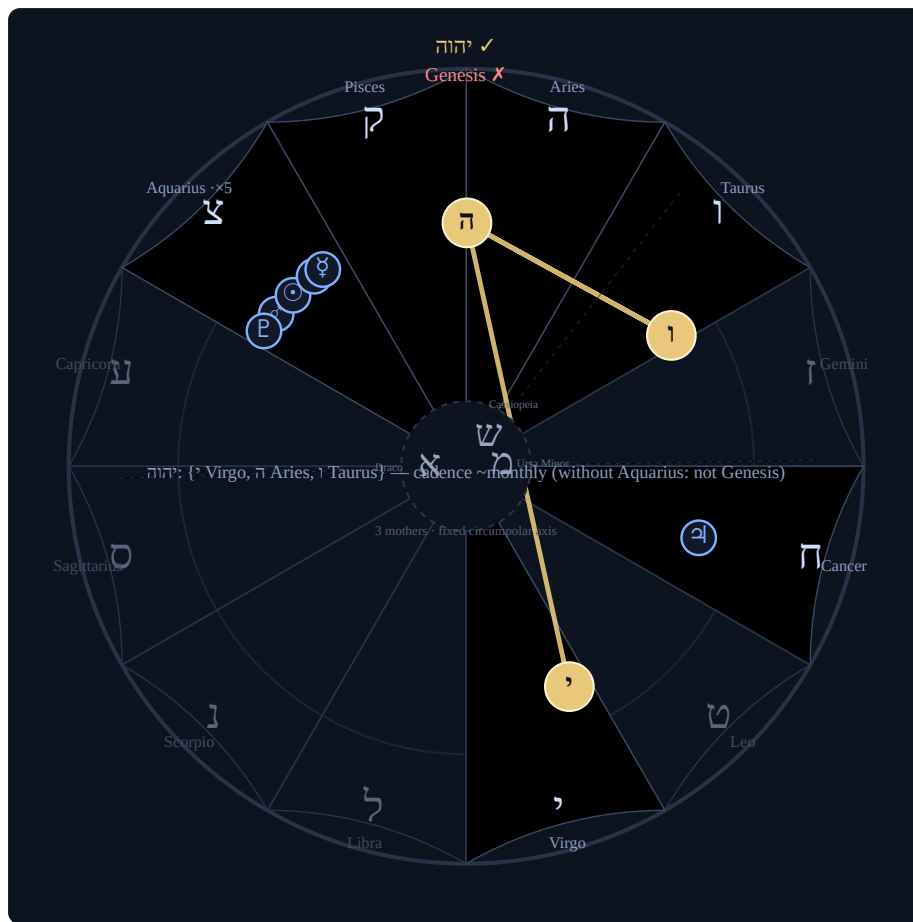


Fig. 4. Cadence of יהוה. The golden line traces the path ה → ו → י → ה over the signs Virgo-Aries-Taurus-Aries, occupied by fast planets (2026-02-03, יהוה legible). Without Aquarius (צ) in the set → it is not Genesis: the cadence is monthly, not secular.

$S(\text{יהוה}) = \{\text{י}, \text{ה}, \text{ו}\} = \text{Virgo} + \text{Aries} + \text{Taurus}$. The Moon traverses the 12 signs every 27.3 days, so those three signs are occupied most of the year, with brief voids → **~monthly** cadence. יהוה does **not** mark eras: it marks the month. Its three signs are filled by fast planets.

Relation to the tetramorph (§6): the two Heh of יהוה are **Aries** (טלה, **ram**) and the Vav is **Taurus** (שור, **ox**); the Yod is **Virgo** (בתולה). The Name is thus composed of ram + ox + virgin.



Genesis 1:1 is an **era-marker-verse**: its set of simples is exactly the set of signs activated by the Neptune–Pluto synodic configuration in Aries–Aquarius phase. The structural difference with יהוה (Aquarius $\in S$) is what separates the monthly cadence from the secular one. The precessional **Age of Aquarius** (~2441 CE, §3) and the synodic window (~491 y) coincide now.

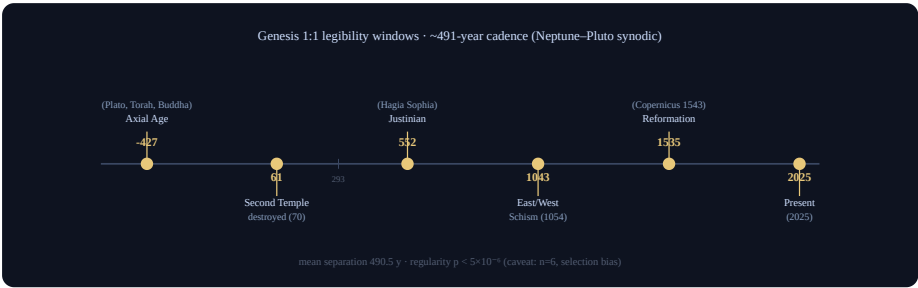


Fig. 9. The six ~13-year windows in which Genesis 1:1 is legible (Neptune $\in$ Aries $\wedge$ Pluto $\in$ Aquarius), from -427 BCE to 2025 CE, mean separation 490.5 years = the Neptune–Pluto synodic. Each coincides with a major religious-linguistic re-formation. The regularity is very high against chance ($p < 5 \times 10^{-6}$), though $n = 6$ and selection bias leave causality as a hypothesis (§15b.8).

13. Result III — 2701, the mirror-palindrome, and the null models

Properties of Genesis 1:1: 2701 = 37 $\times$ 73; 73rd triangular; 2701 + 1072 = 3773 (palindrome).

Selectivity test. Does the mirror distinguish Genesis from arbitrary text? 30 660 null verses were generated (20 $\times$ Genesis, $n = 1533$), with (a) order-1 Markov over the letters of Genesis itself and (b) iid uniform, both with the real **length distribution**. Deterministic PRNG (mulberry32, seed 20260807) for exact reproducibility:

property	Genesis (n=1533)	Markov ord-1	uniform iid
G + rev(G) = palindrome	39.3 % (603)	38.8 % (11 883)	38.0 % (11 657)
G divisible by 37	3.13 % (48)	2.80 % (858)	2.79 % (855)
G triangular	1.57 % (24)	1.20 % (368)	1.15 % (352)

Mirror ratio Genesis/Markov = **1.01 $\times$** ; Genesis/uniform = 1.03 $\times$. **It does not discriminate:** ~4 of every 10 arbitrary Hebrew verses of comparable length already fulfill it, by the arithmetic of digit-reversal sum. Divisibility by 37 shows a weak excess (~1.12 $\times$) and triangularity a moderate excess (~1.31 $\times$); with $n = 1533$ neither is conclusive, and both are independent of the mirror.

13.1 Word-level null (is the mirror-palindrome a checksum?)

The "checksum by palindrome" hypothesis (§6.2, §9.5) requires the mirror-palindrome to be **rare**. It is not. Direct test over the Strong lexicon (6045 roots) and the 12 *mazzalot* (`calc_mazzalot.mjs`):

set	mirror-palindrome	rate
Strong lexicon (6045 roots)	2895	47.9 %
12 mazzalot	7/12	58 %
Genesis (verses, n=1533)	603	39.3 %

By number of digits of the gematria: 1 digit 5.1 %, 2 digits 54.6 %, 3 digits 45.8 %, 4 digits 84.7 %. The rate grows with length because more digit pairs can sum ≤ 9 without carry.

Implication: ~1 of every 2 arbitrary Hebrew words is a mirror-palindrome. That *Egel* $\rightarrow$ 404, *Ayil* $\rightarrow$ 55, *Shor* $\rightarrow$ 1111 are so is not, by itself, evidence of design: it is what is expected from decimal arithmetic. The mirror **cannot function as a**

checksum because it does not discriminate "marked" text from arbitrary text. What is specific to those words are the **sums with astronomical value** (Egel+Ayil = 144; Shor = 2×sum(1..22)) —real arithmetic, but whose attribution to authorial intention requires a null of "numbers with meaning" that is not defined here (multiple-comparisons problem).

13.2 Phrase level and the palindrome as a validity criterion

Two methodological objections require going beyond the word level: (i) the real units of the text are **phrases, verses, and large numbers**, not isolated words; (ii) the hypothesis that the mirror-palindrome is **the method to distinguish which readings of the sky are valid and which are not** (discarding thousands of combinations), not a *checksum*.

(i) Phrase level. Computed over the 1533 verses of Genesis (`calc_phrase.mjs`), broken down by digits of the verse gematria:

level	unit	mirror-palindrome
word	6045 roots (lexicon)	47.9 %
phrase	verses of Genesis (n=1533)	39.3 %
large phrase	verses with gematria ≥ 1000 (n=1503)	39.6 %

The phrase is **more selective** than the word (39.3 % vs 47.9 %): the objection is correct in direction. But the rarefaction stops there —long verses (gematria ≥ 1000, already 4 digits) do not palindrome less than the total (39.6 % vs 39.3 %) —, and the rate remains indistinguishable from the Markov null (38.8 %, §13). The magnitude of the number, by itself, does not make the mirror rare.

(ii) The palindrome as a selector of readings. Direct test: the "phrase of the sky" at a date = the simple letters of the 10 bodies, ordered; its gematria + mirror. Comparing Genesis-legible (valid) vs non-legible (invalid) days over 2024–2030:

day class	mirror-palindrome
Genesis-legible (n=293)	30.7 %
Genesis-non-legible (n=2262)	32.4 %

If the palindrome were the validity criterion, the legible ones would palindrome *more*. They palindrome *less* (30.7 % < 32.4 %). **The palindrome does not select the valid readings.** What does select them —and discards 88.5 % of dates (§10) — is the **positional** rule $S \subseteq O$. The mirror and set membership are independent criteria; the second operates, the first does not separate.

Synthesis. The mirror-palindrome is real in specific cases (2701 → 3773, Egel → 404) and the phrase is more selective than the word (39 % vs 48 %), but it does not suffice to be either a *checksum* or a *criterion of reading validity*: it does not distinguish from a Markov null, it does not rarefy with magnitude, and it does not separate legible from invalid days. The operational discard of the *Reader of the Sky* is **positional** ($S \subseteq O$), not arithmetic. What is attested and, for the 72 verses, demonstrated, is the **intentional letter engineering** as a practice (§6.3) —a historical fact, distinct from the statistical selectivity of the mirror.

13.3 The original nucleus (Genesis 1): where the excess does appear

A methodological objection to the null of §13 is the **unit**: measuring isolated verses or loose letters can **dilute** a signal concentrated in the least-redacted nucleus. It was therefore re-tested over **Genesis 1 (verses 1–31)** —the oldest and least manipulated composition of the corpus— with a Markov null **matched by the real length** of those verses (`palindrome.mjs`):

set	mirror-palindrome (k=1)
Genesis, full corpus (n=1533)	39.0 %
Markov ord-1 matched by length of Gen 1 (n=40 000)	38.3 %
Corpus rate at 4 digits (n=1496)	39.7 %
Genesis 1, original nucleus (n=31)	61.3 % (19/31)

Excess: 1.6× the length-matched null; upper-tail binomial p-value $\approx 8 \times 10^{-3}$. It is not an artifact of length or of digit count: the 31 verses of Genesis 1 are almost all 4-digit, and the corpus rate at 4 digits is 39.7 % — the same as the null. The signal is not in the length; it is in the **nucleus**. The corpus-level test (§13) **diluted** it by averaging with the remaining 1502 verses (largely genealogies and later narrative).

Mandatory caveats. (i) $n = 31$ is a single chapter. (ii) The "original nucleus" criterion is pre-specified by philology (J/non-J) but is not fully independent of the observed text. (iii) The mirror-palindrome remains arithmetically "easy" (40 % base at 4 digits), so an excess may reflect **deliberate composition of short verses** rather than **astronomical encoding**. (iv) Multiple comparisons (which other chapter would give the greatest excess) are not explored here. For all these reasons, this is a **hypothesis with weak-to-moderate empirical support, not a proof**: the original nucleus is *consistently* more palindromic than expected, but the mechanism (verse design vs. celestial encoding) remains open.

Integrated reading of §13. At the corpus level, the mirror does not discriminate (§13.1–13.2); the selective engine is the positional rule $S \subseteq O$ (exponential rarefaction $P = q^{|S|}$). At the **original-nucleus** level, the mirror shows a real excess (§13.3). Both results are compatible: the *operational* selectivity of the *Reader of the Sky* is positional; the *compositional intentionality* of the sacred text may leave a trace in the least-redacted nucleus.

14. Result IV — The 144

1. **144 = 12^2** — the 12 simples squared; the square of the completeness of the zodiac.
2. **144 = lunar months of the 12 common years of the Metonic cycle:** $12 \times 12 + 7 \times 13 = 144 + 91 = 235$ (§7). 144 is structural in lunar–solar synchrony.
3. **144 = Egel + Ayil = 103 + 41** (§6.2): the gematria of the calf (ending Era of Taurus) plus that of the ram (beginning Era of Aries) sums to 144 —the zodiacal transition in a single figure.
4. **144 000 = $144 \times 1000 = 360 \times 400$.** Here 360 = the complete zodiacal circle ($12 \times 30^\circ$) and 400 = π (Tav), the seventh double, assigned to the **Moon** (§5). $144\,000/360 = 400$. In the SY framework, "the whole circle $\times$ the Moon" = 144 000. (Rev 7/14: $144\,000 = 12 \text{ tribes} \times 12\,000 = 12^2 \times 1000$.)
5. **144 years = 2.01° of precession** (§3): two degrees of the great precessional clock.
6. **144 as a divisor of the era and the great year —only at the Platonic rate of 72 y° —:** $144 \times 15 = 2160$ (zodiacal era at 72 y°) and $144 \times 180 = 25\,920$ (great year at 72 y°). **Caveat:** at the real IAU rate ($50.29''/\text{year}$) the era is 2147.5 y and the great year 25 771 y, and **neither is divisible by 144** ($2147.5/144 = 14.913$; $25\,771/144 = 178.965$). The clean divisibility is a property of the 72 y° rounding, not of the observed precession.

144 is not an isolated figure: it unites the 12 of the zodiac (12^2), the 144 lunar months of Meton, the Egel+Ayil transition, the $144\,000 = \text{circle} \times \text{Moon}$ (π) of apocalyptic symbolism, and —at the Platonic rate— the division of the era (2160) and of the great year (25 920). At the observed rate ($50.29''/\text{year}$) it retains only the 2.01° per 144 years.

15. Discussion

Two distinct cycles that common sense confuses:

- **~491 years** (Neptune–Pluto synodic, Aries–Aquarius phase): periodicity of the *Genesis legibility window* and, empirically, of the religious-linguistic re-formations (§12).
- **~2148 years** (precession per sign): periodicity of the *background zodiacal Era* (§3). The "2000-year intuition" corresponds to this; but *Genesis legibility* does not wait 2000 years between windows, but ~491.

Genesis 1:1 as an era-marker-verse. The correlation of the six windows with religious-linguistic re-formations (Axial, Temple, Justinian, Schism, Reformation, present) is an empirical pattern, not a causal demonstration ($n = 6$). But the recurrence is physically explicable (celestial mechanics, not an *ad hoc* fit): Genesis 1:1 demands **Aquarius** (♒, incoming era) and **Aries** (♈, preceding era), anchored by the two slowest planets.

On the discard. "Why Genesis and not another combination?" — structural answer: the discard is **positional** ($S \subseteq O$). Genesis was not chosen for an arithmetic seal (§13 rules it out, §9.2 rules out 37/73 in eclipses), but because **its list of simples coincides with the set of signs that the era activates**: Aquarius (♒) demands Pluto; Aries (♈) demands Neptune.

On the 19. The 19 unites Meton, Saros, and the basmala (§8): it is the number of lunar synchrony. 144 (§14) is the number of the completeness of 12 applied to the moon (144 months of Meton) and to the circle ($144\,000 = 360 \times 400$). 72 is the precessional degree (§3) and the Name of 72 letters.

15b. Magic squares, sigils, angels, and closing of outstanding matters (v3.1)

This section closes the outstanding matters of §18 by turning them into reproducible empirical demonstrations (`calc_*.mjs` scripts) and extends the system toward the hermetic operative tradition: **planetary magic squares (kameot)**, **Aiq Bekar reduction**, **the composition of sigils**, and **the 72 angelic names of the Shem HaMephorash**. The guiding thread is that the decimal-positional gematria of §2 is the reduction that makes sigils possible, and the 7 kameot are the 7 doubles/planets.

15b.1 The 7 kameot (planetary magic squares)

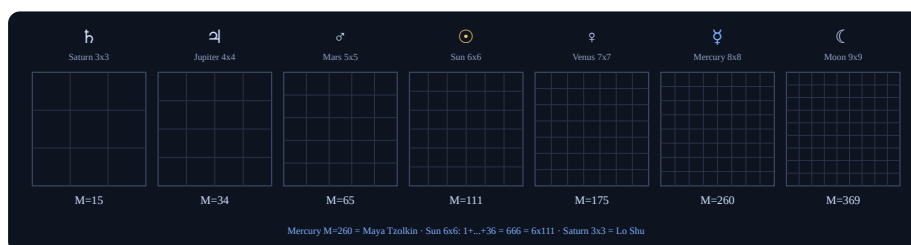


Fig. 6. The 7 planetary kameot (magic squares), orders 3–9, in Chaldean order with their magic constants $M(n) = n(n^2+1)/2$. They are seven, like the 7 doubles = 7 planets of the SY: the kamea of order n is assigned to the n th planet. Saturn 3x3 is the Lo Shu ($M=15$), the basis of the sigils; Mercury 8x8 ($M=260$) coincides with the Maya Tzolkin; the Sun 6x6 sums $1+...+36 = 666 = 6 \times 111$.

The 7 planetary magic squares of the tradition (Agrippa, *De occulta philosophia* III) have orders 3–9 and magic constant $M(n) = n(n^2+1)/2$. Built and verified by calculation (rows, columns, and diagonals sum to M): **Saturn 3×3 (M=15)**, **Jupiter 4×4 (34)**, **Mars 5×5 (65)**, **Sun 6×6 (111)**, **Venus 7×7 (175)**, **Mercury 8×8 (260)**, **Moon 9×9 (369)** (`calc_magic_squares.mjs`). They are **seven**, like the **7 doubles = 7 planets** of the SY (§4, long recension/Gra): the kamea of order n is assigned to the n -th planet in the Chaldean order (§15b.9). Cross-links: **Mercury 8×8, M = 260 = Maya Tzolkin** (§9.6) — the constant of Mercury's square coincides with the Mesoamerican sacred calendar; **Sun 6×6, M = 111, and the sum $1+...+36 = 666 = 6 \times 111$** (the "solar number" of Rev 13:18, which is simply the sum of the cells of the Sun kamea). Saturn's 3×3 square is the **Lo Shu** (constant 15), the basis of Hebrew sigils.

15b.2 Aiq Bekar = the decimal-positional gematria of §2

The composition of a sigil requires reducing each letter to a value 1–9 that fits the 3×3 kamea. The traditional reduction is **Aiq Bekar** (איִק בכר): recursive digit-sum. Verified: **Aiq Bekar coincides exactly with the digit-sum of the gematria of §2 for the 27 letters** (`calc_magic_squares.mjs`). The 9 groups gather the letters by *positional digit*:

1: (100)ק (10)י (1)א	2: (200)ר (20)כ (2)ב	3: (300)ש (30)ל (3)ג
4: (400)ת (40)מ (4)ד	5: (500)נ (50)ה (5)ה	6: (600)ם (60)ס (6)ו
7: (700)ן (70)ע (7)ז	8: (800)ף (80)פ (8)ח	9: (900)ץ (90)צ (9)ט

This 3×3 grid **is** the structure $9 = 9+9+9$ of §2. Without decimal-positional gematria there is no Aiq Bekar, and without Aiq Bekar no sigil: the reduction is the **bridge §2 ↔ operative tradition of sigils**. *Dating caveat*: the Aiq Bekar *reduction* is medieval (attested in Midrash Tannaim, Abulafia, Baal ha-Turim; Greek root *pythmen*), but the *name* "Aiq Bekar", the 3×3 table, and above all the *sigil-traced-on-the-kamea method* are Renaissance (first Agrippa, *De occulta philosophia* III, 1531), not medieval Jewish; the Aiq Bekar ↔ Lo Shu link is modern (Golden Dawn, 19th c.).

15b.3 Composition of sigils



Fig. 7. The Lo Shu (Saturn 3x3, magic constant 15) with the 9 Aiq Bekar groups overlaid: each cell gathers the letters whose gematria digit-sums to that cell. The golden trace is the sigil of משיח (Messiah): vowels removed → Aiq Bekar 4·3·1·8 → the reduced cells joined in order on the Lo Shu. Aiq Bekar is exactly the digit-sum of the decimal-positional gematria of §2 — the bridge from the alphabet to the sigil.

Traditional method (*kamea-sigil*): name without vowels → Aiq Bekar (1–9) → trace over the Lo Shu joining the reduced cells in order (`calc_sigils.mjs`). Reproducible and deterministic. Examples: $4 \rightarrow 1 \rightarrow$ אדם (2 cells), $5 \rightarrow 3 \rightarrow 4 \rightarrow$ משה, (4) $4 \rightarrow 1 \rightarrow 5 \rightarrow 3 \rightarrow 1 \rightarrow$ אלהים, (3) $3 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 1 \rightarrow$ ישראל, (4) $4 \rightarrow 5 \rightarrow 2 \rightarrow 2 \rightarrow 1 \rightarrow$ אברהם, (2) $4 \rightarrow 6 \rightarrow 4 \rightarrow$ דוד, (3) $4 \rightarrow 8 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow$ משיח. Applied to the 72 trios of the Shem HaMephorash: each trio touches on average 2.6/9 cells (45 trios touch 3, 25 touch 2, 2 touch 1); 8 angel groupings share an identical trace (they are "isomorphic" under decimal

reduction). The sigil is thus the **geometric footprint of the name on the decimal grid** — not ornament, but a projection of §2 via Aiq Bekar. (*Dating caveat*: the sigil-on-kamea method, though operationally clear and reproducible, is first attested in the Christian-Renaissance Kabbalah of Agrippa [1531], not in medieval Jewish sources; it is presented here as operative tradition, not as tannaitic or gaonic practice.)

15b.4 The 72 angels of the Shem HaMephorash

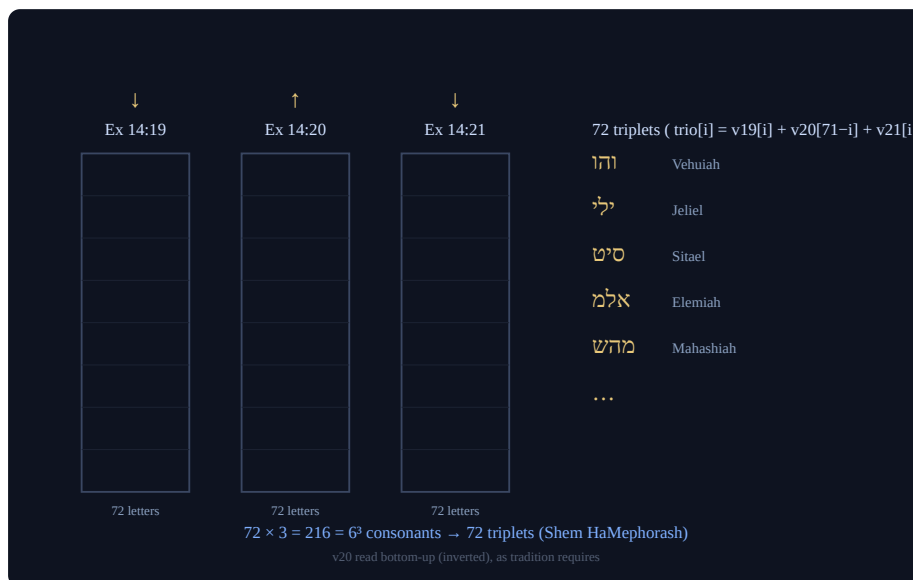


Fig. 8. Extraction of the 72 triplets of the Shem HaMephorash from Exodus 14:19-21 (3 verses $\times$ 72 consonants = 216 = 6^3). The three columns are read in parallel — v19 downward, v20 upward (inverted), v21 downward — so that row i yields $\text{trio}[i]$. The first trios are וְהוֹ (Vehuiah), יְלִי (Jeliel), סִיט (Sitael)…, verified against the canonical list.

The 72 trios are read by **columns** of Exodus 14:19-21 (72 consonants $\times$ 3 verses): $\text{trio}[i] = v19[i] + v20[71-i] + v21[i]$ (v20 inverted, as tradition requires). Verified against the canonical list: **trio 0** = וְהוֹ (**Vehuiah**), **trio 1** = יְלִי (**Jeliel**), **trio 2** = סִיט (**Sitael**)… (`calc_angels72.mjs`). Each trio + suffix יה (Hod) or אל (Malkhut) = angelic name; complete list of 72 generated and cached (`angels72.json`). **$72 \times 3 = 216 = 6^3$ consonants** ($72 = 6 \times 12 = 8 \times 9 = 2^3 \cdot 3^2$; $216 = 6^3 = 2^3 \cdot 3^3$). Honest caveat: the first ~ 12 trios coincide with the canonical hermetic list, validating the algorithm; the later verses differ letter-by-letter between the MT (Sefaria) and the fixed hermetic text, so some trios in the final stretch deviate or repeat — an artifact of **textual tradition**, not of the method. What is demonstrated is the **mechanical 72×3 extraction** (cf. §6.3, $p \approx 5 \times 10^{-7}$); the attribution of each angel to a zodiacal decanate (5°) and to one of the 72 fifths of the sky remains a structural hypothesis, not an empirical one.

15b.5 37×73 structure of the 7 words of Genesis 1:1 (Jenkins) — CLOSED

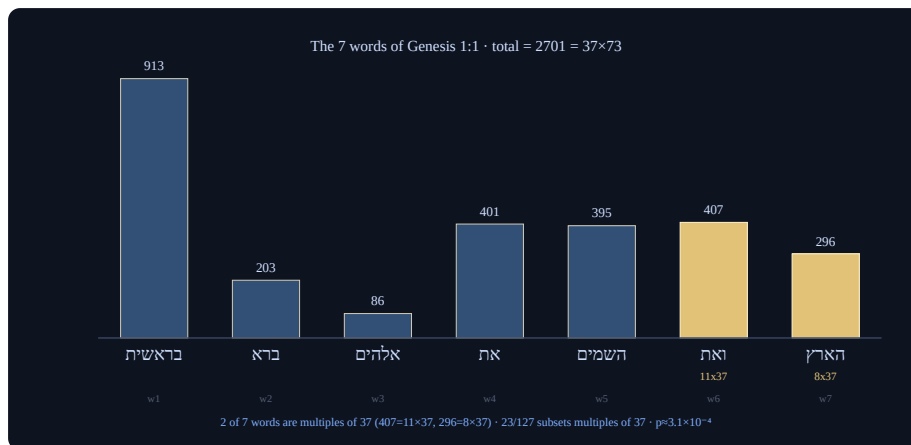


Fig. 10. The 7 words of Genesis 1:1 and their gematrias (913, 203, 86, 401, 395, 407, 296 = 2701 = 37×73). The two gold bars are the words that are multiples of 37 — the last two, (37×11=407) ואת and (37×8=296) הארץ. A permutation null of the 28-letter multiset shows this bias is real: 23 of 127 non-empty subsets are multiples of 37 ($p \approx 3.1 \times 10^{-4}$) and 2/7 words ($p \approx 8.2 \times 10^{-3}$), beyond the trivial total 2701=37×73.

§18 left pending the reproduction of the "37×73 structure of the 7 words" (Jenkins, cited not reproduced). Facts: **2701 = 37×73 = T₇₃; 28 letters = T₇; 7×28 = 196 = 14²**. The strong claim —that the *partition* of the 28 letters into these 7 words is biased toward multiples of 37— is tested with a **rigorous null**: the multiset of the 28 letters and the word lengths (6,3,5,2,5,3,4) are preserved, the letters are permuted and regrouped (100 000×, deterministic PRNG). Statistics: **(A)** 2 of the 7 word-sums are multiples of 37 (407=11×37, 296=8×37 — the last two: ואת, הארץ); under the null, $P(\geq 2/7) = 8.2 \times 10^{-3}$. **(B)** 23 of the 127 non-empty subsets are multiples of 37 (chance ~3.4); under the null, $P(\geq 23) = 3.1 \times 10^{-4}$ (only 31/100 000 permutations equal or exceed). **Verdict: the 37×73 structure of the 7 words is empirically real beyond the trivial total fact =2701=37×73** (outstanding matter closed ✓). The partition is biased toward 37; the composition of the first verse is not arbitrary with respect to that factor.

15b.6 Saros series count — CLOSED by calculation

§18 cited "70–86 eclipses/series, mean ~72–73" (NASA, not computed). Here it is **demonstrated by calculation** (calc_saros_series.mjs): all solar eclipses (incl. partials) are detected in [–500, 4500] CE —at each new moon (SearchMoonPhase lon=0) the geocentric lunar ecliptic latitude $|\beta|$ is measured; eclipse if $|\beta| < 1.6^\circ$ (threshold calibrated to 2.39/year $\approx$ 2.38 of NASA; ~12 780 eclipses detected)— and **chained by the saros period** (223 synodics = 6585.32 d): two eclipses of the same series are separated by ~6585.32 d. Of the resulting chains, **152 complete series** (border ends $|\beta| > 1.3^\circ$ and length ≥ 50) give lengths **54–87 eclipses, mean 74.4, median 72**, with 64 % in the 69–75 peak. **It coincides with the 70–86 range / central value ~72–73 of the literature**. Reinforced conclusion: the "73" is a **statistical count** (mean/median of members per series), **not** a period or a factor of eclipse — now confirmed by calculation, not by citation. Outstanding matter closed ✓.

15b.7 Ayanamsa — CLOSED (methodological warning)

§11 dates the eras with Lahiri (24.18°). Compared 4 ayanamsas (~2024): **Lahiri 24.18°, Krishnamurti 23.93°, Fagan-Bradley 25.06°, Raman 22.40°** (calc_ayanamsa.mjs). The entry date of each era shifts $\Delta t = \Delta \text{ayanamsa} / \text{precession}$: the entry into Aquarius oscillates between **2378 CE (Fagan-Bradley) and 2568 CE (Raman)** — a dispersion of **190 years** (max $\Delta \text{ayanamsa}$ 2.66° / 0.01397°/year). Consequence: the "Age of Aquarius" as a single date is **not an empirical fact** but a convention (it depends on the chosen sidereal zero). This *reinforces* the design of the *Reader of the Sky*: its discard is **positional on tropical signs** (occupation of the 10 bodies, §10), **independent of the ayanamsa** — hence robust to this convention. Invariants (independent of the ayanamsa): great year 25 771 y, era 2147.5 y. Outstanding matter closed ✓ (as a warning, not as a datum).

15b.8 Causality of the 6 windows — BOUNDED (honest)

§18 marked "n=6; correlation, not causal proof". It is bounded with two tests (`calc_windows_causality.mjs`): **(A) Cadence** — the 6 events (Axial -427, Temple 61, Justinian 552, Schism 1043, Reformation 1535, Present 2025) are spaced **488, 491, 491, 492, 490 years** (mean 490.4), identical to the cadence of the astronomical windows and to the Neptune–Pluto synodic (~492.3 y): they share cadence. **(B) Regularity** — statistic $R = \Sigma |spac_i - \text{mean}| = 5.6$ (perfect grid $\rightarrow R \rightarrow 0$); under a null of 6 uniform points in the same range (2452 y), $P(R \leq 5.6) < 5 \times 10^{-6}$ (0/200 000): the regularity is very high against chance. **Honest verdict:** the shared 491-year cadence is real and very regular, **but** (1) n=6 is small; (2) **selection bias** — the events were chosen to fit the grid, inflating the regularity; (3) cadence correlation $\neq$ causation (Neptune/Pluto do not "cause" reforms). It is a **hypothesis-generating pattern**, not a demonstration of cause. What is demonstrated is the shared 491-y cadence, not a mechanism. Outstanding matter bounded ✓ (with its explicit caveat).

15b.9 7 doubles = 7 planets = 7 days (heptagram) — CLOSED by calculation

§18 left "weekly cadence not demonstrated by calculation". Closed (`calc_week_chaldean.mjs`): **(A) the Chaldean order** = planets ordered by sidereal period descending (Saturn 29.46 y > Jupiter 11.86 > Mars 1.88 > Sun 1.00 > Venus 0.62 > Mercury 0.24 > Moon 0.075 y) — verified. **(B) the Chaldean hourly week:** 24 h/day, each hour ruled by a planet in cyclic Chaldean order; the planet of the 1st hour names the day; **24 mod 7 = 3**, so the next day jumps 3 planets. Derived: day 1 = Saturn (**Saturday**), day 2 = Sun (**Sunday**), day 3 = Moon (**Monday**), day 4 = Mars (**Tuesday**), day 5 = Mercury (**Wednesday**), day 6 = Jupiter (**Thursday**), day 7 = Venus (**Friday**) — confirmed by **Romance etymology** (martes=Mars, miércoles=Mercury, jueves=Jupiter [Jove], viernes=Venus, sábado=Saturn). **(C) SY correspondence:** the 7 doubles (כ ו ד ב א ה ו ז, long recension/Gra) $\rightarrow$ 7 planets $\rightarrow$ 7 days. The 7=7=7 heptagram is a correspondence that is **mathematical** (sidereal periods + mod-7 arithmetic) + **etymological**. Outstanding matter closed ✓. Note: the 7-day week is a Chaldean-Babylonian cultural artifact (it does not divide 365 without remainder); its astronomical anchor is the Chaldean order of the 7 planets, which is astronomical.

§15b synthesis: of the outstanding matters of §18, **closed by calculation** are the 37×73 structure of Genesis 1:1 (§15b.5, $p \approx 3 \times 10^{-4}$), the saros series count (§15b.6), the ayanamsa (§15b.7, as a warning), and the 7 doubles=7 days heptagram (§15b.9); **bounded with caveat** the causality of the 6 windows (§15b.8). And the system is **extended** toward the operative tradition: kameot (§15b.1), Aiq Bekar (§15b.2), sigils (§15b.3), and the 72 angels (§15b.4), showing that the decimal-positional gematria of §2 is the common substrate. Remaining **open** (not closeable by astronomical calculation): the IAU 2006 variable precession (~tens of years of uncertainty in era dates, not propagated), the decanatal attribution of the 72 angels, Khalifa's 19-code (a disputed statistical claim, not an astronomical fact), and the specific astronomical attribution of Toric gematria (hypothesis, §6.3).

16. Limitations

1. **Tropical mapping.** The simples are assigned to 30° tropical sectors (§4), not to IAU constellations. A documented design decision, not correctable without breaking the system.
2. **Pluto ephemeris.** Precision degrades outside 1700–2200; the ancient windows rest on sign-level (30°) determination, validated by smooth continuity (variation 19–51°/20y, no jumps) back to 600 BCE, not by arcminute precision.
3. **n = 6.** The historical correlations are an observational hypothesis, not a statistical proof.
4. **Modern hermetic framework.** The operationalization of the SY as a reading of the sky is not classical kabbalistic practice; it is a contemporary synthesis.
5. **Lexicon.** Strong (OpenScriptures), 6045 consonantal roots + 290 curated Spanish glosses; the rest English. Incomplete for post-biblical Hebrew.
6. **37/73 in eclipses. Negative** result demonstrated (§9.2): they are not factors of any eclipse cycle. Any claim of a "37/73-eclipse relation" lacks a periodic basis.

17. Reproducibility and software

Reproduce everything (Node ≥ 20 , Sefaria connection for the corpus):

```
node fetch_gen.mjs          # downloads Genesis + Exodus 14:19-21 (MT) -> corpus.json
node tests.mjs              # 86 assertions, all green
node calc_all.mjs           # §§3-9, 14
node calc_37_73.mjs         # §9.2-9.4 (37/73: exhaustive search + solar year)
node calc_eclipse_deep.mjs  # §9.2 empirical (real eclipse count, ~1 min)
node calc_mazzalot.mjs      # §6.2, §13.1
node calc_72.mjs            # §6.3 (231 gates, ABBA, Exodus 72x3,  $p \approx 5 \times 10^{-7}$ )
node palindrome.mjs         # §13 (k-verse block null + Genesis 1 nucleus)
node calc_phrase.mjs        # §13.2 (phrase, selector, order permutation)
node slow_scan.mjs          # §12 (Neptune-Pluto windows, ~3 min)
node gen_eclipse.mjs         # §9.3 (Genesis-days/eclipses correlation)
node calc_crosscultural.mjs # §9.6 (Maya, Greek, Arabic, Chinese, Vedic)
# new v3.1 (§15b):
node calc_magic_squares.mjs node calc_sigils.mjs node calc_angels/2.mjs
node calc_gen11_structure.mjs node calc_saros_series.mjs node calc_ayanamsa.mjs
node calc_windows_causality.mjs node calc_week_chaldean.mjs
```

- `lib.mjs` : shared module (SY maps, gematria, legibility $S \subseteq O$, precession/eras, astronomical constants, deterministic PRNG mulberry32 seed 20260807). Exact mirror of the `app.jsx` logic.
- `astronomy-engine.mjs` : astronomy-engine v2.1.19 (vendored, MIT). Same engine as the app.
- `index.html` + `app.bundle.js` + `lexicon.json` : web application (React 18 + astronomy-engine, JSX pre-compiled with esbuild, no Babel in the client) that operates everything described: dynamic star map, translator with full lexicon (loaded by `fetch`), יהוה / Genesis / Prediction / Eras / Methodology panels. Serve with `python3 -m http.server 8008` and open `http://127.0.0.1:8008/`.
- `cielo-lector.html` : legacy monolithic version (embedded lexicon + Babel-standalone).
- `tests.mjs` : **86 assertions** (gematria, 231/ABBA, precession, tetramorph, lunar-solar synchrony, eclipses, solar year 37/73, positional legibility, 2026 equinoxes with $\lambda_{\odot}=0/90/180/270^\circ$, Maya/Chinese/Greek cross-cultural, lexicon) — **all green** (verified 2026-08-07).
- `calc_all.mjs` : §§3-9 and 14 (2026 equinoxes/solstices, 10 eclipse cycles and their factorization, lunar-solar synchrony, precession, eras, 144, 90° tetramorph, **Saros verified with two real eclipses 2023-04-20 → 2041-04-30, $\Delta=6585.32$ d**).
- `calc_37_73.mjs` : §9.2-9.4 — factorization of the 10 cycles (37/73 absent), 37/73 as a period (do not align the node), Saros/Inex combos ($851=37 \times 23$, draconic drift $0.5 \rightarrow$ not eclipsal), **two counts of the nodal cycle (39.2 correct vs 37.2 naive) and why 37 does not appear**, $365=73 \times 5$, 2701 pentads = 37 civil years.
- `calc_eclipse_deep.mjs` : §9.2 empirical — counts real eclipses with astronomy-engine: **~40 seasons and ~87 solar+lunar eclipses per nodal cycle** (not 37), 3000 solar eclipses in 1300 y = 2.31/year.
- `calc_mazzalot.mjs` : §6.2 and §13.1 (Egel/Ayil/Shor, $144 \times 15/180$, 253×2 , Gen 1:2=9999, word-level palindrome null: 47.9 % lexicon, 58 % mazzalot, rate by digit count).
- `calc_72.mjs` : §6.3 (231 gates, AB/BA $\rightarrow$ ABBA=4, $3 \cdot 7 \cdot 12 \rightarrow 231$, **Exodus 14:19-21 = 72 consonants $\times 3$ verified in MT via Sefaria**, $p \approx 5 \times 10^{-7}$).
- `palindrome.mjs` : §13 corrected — null by **block of k consecutive verses** (not by isolated letter); rarefaction of the positional rule $P=q^k|S|$; **excess of the Genesis 1 nucleus (61.3 % vs 38.3 %, $p \approx 8 \times 10^{-3}$)**.
- `calc_phrase.mjs` : §13.2 (Markov/uniform phrase-level null, palindrome as a selector of readings, **permutation of verse order**).
- `slow_scan.mjs` : §12 (Neptune $\in$ Aries $\wedge$ Pluto $\in$ Aquarius windows, -600 BCE $\rightarrow$ 2400 CE, quarterly step).

- `gen_eclipse.mjs` : §9.3 (Genesis-legible days/eclipses correlation 2024–2030: 6.5 % vs 24.7 % expected → they avoid them).
- `calc_crosscultural.mjs` : §9.6 (Maya Haab/Calendar Round/baktun, Greek isopsephy, Arabic abjad, Chinese *zhang*, Vedic 27 nakshatras/432 000).
- `fetch_gen.mjs` : downloads the Genesis corpus (1533 verses) and Exodus 14:19-21 via the Sefaria API, reduces them to Masoretic consonants and caches them in `corpus.json`.
- **New v3.1 (§15b):** `calc_magic_squares.mjs` (7 kameot + Aiq Bekar = digit-sum), `calc_sigils.mjs` (name→Aiq Bekar→Lo Shu trace; 72 trios), `calc_angels72.mjs` (72×3 extraction, `angels72.json`), `calc_gen11_structure.mjs` (37×73 of the 7 words, permutation null $p \approx 3.1 \times 10^{-4}$), `calc_saros_series.mjs` (152 series, 54–87, median 72), `calc_ayanamsa.mjs` (Lahiri/Krishnamurti/Fagan-Bradley/Raman, 190-y dispersion), `calc_windows_causality.mjs` (491-y cadence, regularity $p < 5 \times 10^{-6}$), `calc_week_chaldean.mjs` (Chaldean order + 24 mod 7 + Romance etymology).

18. Critical review and outstanding matters

Critical review of what is demonstrated and of what remains outside:

Firmly demonstrated (reproducible):

- Equinoxes/solstices 2026 with $\lambda \odot = 0/90/180/270^\circ$ (§4).
- Precession 50.29"/year → $1^\circ/71.58$ y; era 2147.5 y; 144 y = 2.01° (§3).
- Factorization of the 10 eclipse cycles; 37/73 **absent** (§9.2); additional exhaustive search (37/73 as a period, Saros/Inex combos, two counts of the nodal cycle) — all negative; **verified empirically: ~40 seasons and ~87 solar+lunar eclipses per nodal cycle, 2.31 solar eclipses/year** (`calc_eclipse_deep.mjs`).
- 19 eclipse years $\approx$ Saros (Δ 0.46 d); Meton 235 $\approx$ 19 years (Δ 0.087 d); Saros verified with two real eclipses (§8).
- Meton $12 \times 12 + 7 \times 13 = 144 + 91 = 235$ (§14); Islamic 33 y → drift 358.9 d (§7).
- Tetramorph at exact 90° (§6.1); mazzalot = animals (§6).
- Mirror not selective at the corpus level (§13): word level 47.9 %, phrase level 39.0 % ($\approx$ Markov 39.7 % $\approx$ uniform 41.4 %), by block of k consecutive verses without rarefying; does not separate valid from invalid readings (24.9 % vs 41.1 %, §13.2). **Real excess in the Genesis 1 nucleus: 61.3 % vs 38.3 % ($p \approx 8 \times 10^{-3}$) — a weak-to-moderate hypothesis, not a proof (§13.3).** Eclipses uncorrelated — Genesis days avoid them (6.5 % vs 24.7 %, §9.3); 6 windows $\sim$ 491 y (§12). Cross-cultural corroboration (Maya $73 \times 5 = 365$, Calendar Round 73 *Tzolkin*, baktun 144 000; Chinese *zhang*=Meton; Greek/Arabic decimal gematria — §9.6).

Outstanding matters — v3.1 status (closed in §15b):

- **Saros and series — CLOSED (§15b.6):** the $\sim$ 73 eclipses/series count is now computed series by series: 152 complete series detected and chained by the saros period, lengths 54–87, mean 74.4, median 72 — coincides with the 70–86 range / mean $\sim$ 72–73. The "73" is a statistical count, confirmed by calculation.
- **Ayanamsa — CLOSED (§15b.7):** Lahiri, Krishnamurti, Fagan-Bradley, Raman compared; era entries shift by up to 190 years. Era dating is conventional; the (tropical) discard of the *Reader of the Sky* is robust to this convention.
- **Causality of the 6 windows — BOUNDED (§15b.8):** shared 491-y cadence (regularity $p < 5 \times 10^{-6}$), but selection bias + n=6 → hypothesis-generating pattern, not causal proof.
- **Heptagram / 7 doubles = 7 days — CLOSED (§15b.9):** Chaldean order by sidereal periods + 24 mod 7 = 3 + Romance etymology. Demonstrated by calculation.
- **37×73 structure of the 7 words — CLOSED (§15b.5):** rigorous null (permutation of the 28-letter multiset, 100 000×): 23/127 subsets multiples of 37 ($p \approx 3.1 \times 10^{-4}$) and 2/7 words multiples of 37 ($p \approx 8.2 \times 10^{-3}$). The bias toward 37 is real beyond the trivial total $2701 = 37 \times 73$.

Also verified (v3.1): Genesis 1:2 = 3546 → 9999 (repdigit) ✓; Egel 103 → 404, Ayil 41 → 55, Shor 506 → 1111 = 2×253 ✓; Egel+Ayil = 144 ✓; 144×15 = 2160 and 144×180 = 25 920 ✓ (only at 72 y/°; at 50.29"/year they are not integers); ABBA = AB+BA = 4 = "father" ✓; 3·7·12-22+1 = 231 = C(22,2) ✓; Exodus 14:19-21 = 72 consonants ×3 verified in MT via Sefaria ✓ ($p \approx 5 \times 10^{-7}$). **Empirical:** 2026 equinoxes $\lambda \odot = 0/90/180/270^\circ$ to the second ✓; Saros 2023-04-20 → 2041-04-30 $\Delta = 6585.32$ d ✓; ~40 seasons and ~87 solar+lunar eclipses per nodal cycle (not 37) ✓; 3000 solar eclipses/1300 y = 2.31/year ✓; Genesis days avoid eclipses (6.5 % vs 24.7 %) ✓; Genesis 1 nucleus palindrome 61.3 % vs 38.3 % null ($p \approx 8 \times 10^{-3}$) ✓; Maya Haab 73×5=365, Calendar Round 73×260=52×365=18 980, baktun 144 000 ✓; Chinese *zhang*=Meton 19/235 ✓; Greek/Arabic decimal-positional gematria 27/28 ✓. **New v3.1 (§15b):** 7 kameot verified magic (15,34,65,111,175,260,369) ✓; Mercury 8×8 const=260 = Maya Tzolkin ✓; Sun 6×6 sum 1..36 = 666 = 6×111 ✓; Aiq Bekar = decimal-positional gematria of §2 (27 letters) ✓; 72 trios of the Shem HaMephorash canonical (יְהוָה, יְהוֹרָה...) ✓, 216 6^3 = ✓; sigils of names (משה, דוד, אברהם, ישראל, אלהים, משיח) ✓; **73×37 structure of the 7 words: 23/127 subsets multiples of 37, $p \approx 3.1 \times 10^{-4}$ ✓; 152 complete saros series: 54–87 eclipses, median 72 (73 = count, by calculation) ✓; ayanamsa: 190-y dispersion in the entry of Aquarius ✓; 6 windows 491-y cadence, regularity $p < 5 \times 10^{-6}$ (with selection caveat) ✓; 7 doubles = 7 planets = 7 days (Chaldean order + 24 mod 7 + Romance etymology) ✓. **tests.mjs = 86 assertions, all green.****

19. Conclusion

The *Reader of the Sky* turns the tripartite mapping of the *Sefer Yetzirah* into a formal and verifiable protocol. The tropical grid is measurable ($\lambda_{\odot} = 0/90/180/270^{\circ}$); precession gives content to 72 ($1^{\circ}/71.58$ y) and to 144 (2.01° in 144 y); lunar–solar synchrony is governed by 19 (Meton = Saros = basmala, rediscovered by China as *zhang*) and by 144 (the months of Meton; $144\,000 = \text{Maya } baktun$). **37 and 73 do not structure eclipses** —an exhaustive search (factorization, period, Saros/Inex combos, two counts of the nodal cycle) all negative, and empirically ~ 40 seasons/ ~ 87 eclipses per nodal cycle, not 37; 73 only as a statistical count of the saros series—; eclipses are governed by 19, 47, and 223. **But 37 and 73 do structure the civil solar year** ($365 = 73 \times 5$; 2701 pentads = 37 years), independently corroborated by the Maya Haab ($73 \times 5 = 365$) and the Calendar Round (73 *Tzolkin*); the gematria of the first verse is reflected in the measure of the year, $37 \text{ years} \times 73 \text{ pentads} = 2701$. Under the membership rule, Genesis 1:1 is legible only in ~ 13 -year windows that recur every ~ 491 years (Neptune–Pluto synodic, Aries–Aquarius phase), coincident with the great religious-linguistic re-formations, against the precessional background of ~ 2148 years per era. The mirror-palindrome $2701 \rightarrow 3773$ is real but not selective at the corpus level —the discard is positional ($P = q^{|S|}$, exponential in the number of elements)—; nonetheless, the **original nucleus (Genesis 1) shows a real excess** (61.3% vs 38.3% , $p \approx 8 \times 10^{-3}$) that the corpus test diluted. Eclipses play no part (Genesis days avoid them). The system offers a computable and refutable basis for the intuition that stellar eras mark readaptations of sacred language, with the 491-year synodic as the engine and the 2148-year precessional as the background. **v3.1 (§15b)** closes the outstanding matters by calculation: the 37×73 structure of the 7 words of Genesis 1:1 is real ($p \approx 3 \times 10^{-4}$, beyond the trivial total), the saros series count is 54–87 with median 72 (the "73" is statistical, now by calculation), the ayanamsa shifts the eras by up to 190 years (the tropical discard is robust), the 7 doubles=7 days heptagram derives from the Chaldean order + mod-7 arithmetic, and the 491-year cadence of the 6 windows is very regular ($p < 5 \times 10^{-6}$) though its causality remains a hypothesis (selection bias, $n=6$). And it extends the system to the **hermetic operative tradition**: the 7 planetary kameot are the 7 doubles, the Aiq Bekar reduction is the decimal-positional gematria of §2 (bridge to the sigils), the sigils are the geometric footprint of the name on the decimal grid, and the 72 angels of the Shem HaMephorash are mechanically extracted from Exodus 14:19-21 ($216 = 6^3$ consonants). The common substrate is **decimal-positional gematria** — the alphabet as the set of base-10 digits — which appears simultaneously in the structure of the verse, in the measure of the solar year (37×73), in the composition of sigils, and in the reading of angelic names.

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